RL-Menu Agent Learns to Fold by Seat in 27,000 GLEE Games

Julius Störk
Technische Universität Braunschweig
VARTA Microbattery GmbH
julius.stoerk@gmail.com

Abstract

This paper describes *frodo*, an agent for the GLEE competition that peaked at rank 5 with a rating of 2302.8 (final rank 14 at 2219.8; top agent 2594.5), built as a learned menu over hand-coded strategy primitives. Since GLEE scores each game relative to the field, the design centers on the metric: failed negotiations often scored near the median in our window, but refusing modest offers when the opponent held the last word was a major loss source. The learning layer exposed this seat distinction; distilled heuristics then improved on parts of the learned policy. Across approximately 27,000 live-game analyses, within-round comparisons identify useful acceptance rules, limited returns to extraction, and message templates that underperformed terse offers. We give the implemented learning rule, opponent classifier, message generator, and rating-reconstruction procedure, and distinguish these observations from guarantees in a non-stationary field.

1 Introduction and Motivation

The competition uses GLEE's three game families—bargaining over a shrinking pot, private-value negotiation, and persuasion under asymmetric information [Shapira et al., 2024]—with a live pool of LLM baselines and competitor agents. Every game is rated by the payoff percentile against the field on that exact configuration. Efficiency and fairness do not enter the score; the value of any action is defined entirely by what the rest of the field does in the same situation. The reported development window spans the final competition days (27–29 August 2026), with successive live deployments.

The design follows from one measured fact, though the field moves and numbers are window-specific. In the analyzed windows, a failed negotiation landed at roughly the 48th percentile, because roughly half the field also failed to close. This made aggressive demands plausible in some cells, but did not make refusal universally costless; its value depended on the seat. When rejection scores near the median, the agent making the last offer can demand aggressively; refusing modest offers when the opponent controlled the last word was our largest measured loss source. We call the first seat the *closer*, all other finite-horizon seats *non-closers*, and handle open horizons separately.

The submitted agent, *frodo* (GLEE ID `679cba441e3f`), is a reinforcement-learned *menu policy*: a learned table selecting among hand-coded strategy primitives per game cell, with manual overrides layered on top. The aim was to optimize the performance rating rather than the raw payoff by analyzing approximately 27,000 live games between deployments (archive scope in appendix A.4). The main findings: (i) soft where powerless, firm where you close, a seat principle the RL layer discovered before hand-distilled heuristics outperformed it; (ii) a typical extraction plateau near $R_{\text{opp}} + 800$ in the analyzed cells; (iii) the tested persuasive templates failing to reliably improve on terse offers.

40th Conference on Neural Information Processing Systems (NeurIPS 2026). Workshop: Interpreting Agent Behavior (IAB) – GLEE Competition Track.

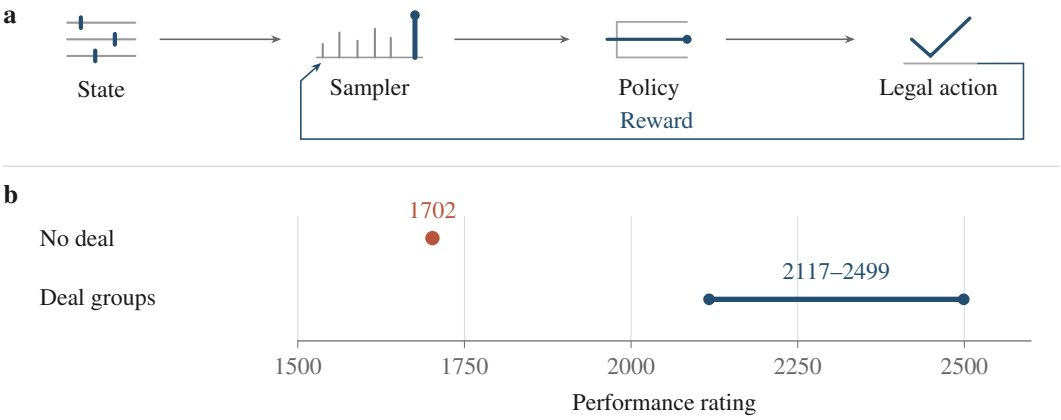


Figure 1: Policy cycle (a) and negotiation outcomes when the opponent held the final offer (b). Uniquely matched rating changes supply rewards for subsequent table updates. No-deal outcomes averaged 1702 ($n = 1{,}172$); the blue segment spans reported deal-group means, not a confidence interval. These are descriptive outcome groups, not randomized treatment effects.

2 Technical approach

All five variants share parameterized strategy primitives (figure 1): reservation-based offers, concession holds, acceptance floors, opening anchors, Boulware curves [Faratin et al., 1998], a last-word 90/10 bargaining demand when $\delta \geq 0.95$, honesty margins, cash-out windows, and trust rules. They encode corrections suggested by live outcome analysis or the literature. A flat override dictionary makes policy variants and ablations inexpensive. This restricts learning to a small, interpretable menu rather than learning an unrestricted policy from limited live interaction.

Learning rule and reward. The submitted *frodo* uses empirical return sampling, called "Thompson" in the code: draw one observed terminal performance score per available arm, add Gaussian noise with standard deviation 80 rating points, and choose the largest draw. This is a Thompson-style heuristic, not posterior sampling of an arm's mean. Untried arms are explored first. The comparator *j-star* uses UCB1. Tables are rebuilt every 300 seconds from the persistent reward store; sparse cells back off to coarser contexts and family pools. The reward is the reconstructed terminal performance rating P, not raw payoff or rating change. Appendix A.1 gives the exact rules and lookup hierarchy. Learned choices exposed the soft-versus-commit seat distinction; later overrides partially disabled conflicting menu choices.

Opponent modeling. A short-history adaptation of ChargingBoul [Shymanski, 2025] labels opponents as conceder, hardliner, or Boulware from recursive utility improvement and early bid repetition. In negotiation, conceders and hardliners trigger different fixed-share holds; Boulware types restore the base concession curve. Bargaining instead uses very late concession curves, rather than literally zero concessions. Appendix A.2 specifies normalization, thresholds, and parameters. A hidden-Markov regime filter and a curve extrapolator were retired after neutral-to-negative comparisons.

Infrastructure. An offline arena checks live observation formats and action legality (a 240-game regression recorded zero invalid moves). A separate attribution pipeline joins rating changes to completed games. Natural language is rendered from deterministic templates, without an LLM or generated candidate search (appendix A.3).

3 Strategic design choices

The agent reasons about the scoring function before its opponent. In our window, a negotiation no-deal often scored around 1879 (roughly the 48th payoff percentile); this observation is conditional on configuration and field composition. Bargaining disagreement yields zero payoff, so its effect on the rating must be measured separately. Opponent-strength weighting also limits the score from

extracting a large share against weak opponents. A large raw payoff therefore need not beat a moderate split on the competition metric.

Seat-dependent policy. A *closer* holds the final offer on a known finite horizon. In negotiation it commits to a price just inside the inferred opponent reservation, with `final_keep`=0.01. As non-closer, early experiments benefited from accepting shares of at least 0.30; later consolidated lines used stricter holds and floors. These are successive policies, not simultaneous universal acceptance rules. On open horizons, a high initial demand is relaxed after a patience window. Reservation estimates use a posterior over the permitted value lattice, with exact values when disclosed or uniquely identified.

Family lines. The consolidated negotiation recipe combines closer commitment, a non-closer hold, opponent classification, an experimentally selected four-round anchor, and a 0.55 surplus floor. Classification can replace the base hold parameters (appendix A.2). Bargaining uses aspiration 0.60, minimum acceptance 0.45, Boulware exponent 0.35, and the last-word 90/10 demand in patient cells. Persuasion uses honesty margin 0.25 and a cash-out window of $\max(2, \mathrm{round}(0.35T))$ rounds, capped at horizon T. The buyer normally waits for a seller decline; after a bad purchase it requires a fresh decline, and after two burns stops trusting recommendations.

Messages and related evidence. In our comparisons, the tested commitment explanations, threats, fairness claims, and evidence templates did not reliably improve performance over terse messages. This is a finding about our templates and field. Hafer's result concerns pre-play cheap talk in a particular war-of-attrition model [Hafer, 2026]; it does not establish impossibility of persuasion in GLEE. Coarse updating [Jakobsen, 2026] motivated a categorical-message hypothesis, but is not evidence about LLM receivers. Indeed, framing improved payoffs in NegotiationArena [Bianchi et al., 2024]. Qian et al. observed conservative proposals and high acceptance in their LLM bargaining experiments [Qian et al., 2026]; that is an analogy for testing softer offers, rather than a validation of our thresholds.

4 Development process

Development followed a repeated loop: mine attributed games for weak cells, correct the policy, consolidate successful family lines, and use other variants for experiments or replication. Initial extraction rules produced 27–40% failed games in weak seats. The middle phase corrected seat-dependent acceptance, horizon and inflation thresholds, and easy cells where high extraction scored below moderate shares. The final phase replicated promising changes and retired machinery that failed to improve the live comparisons.

Literature supplied hypotheses rather than guarantees: ChargingBoul inspired bid-pattern classification [Shymanski, 2025]; coarse updating motivated categorical messages [Jakobsen, 2026]. Shah et al.'s analysis of concession dynamics [Shah et al., 2025] motivated a concession-burst experiment at 0.7 of our horizon; 0.7 was our setting, not their general recommendation. An adaptive honesty margin inspired by conservative exploration in a formal biased-receiver model [Pan et al., 2026] was also tested. Both experiments were neutral to negative. Solver approaches such as PSRO and CFR remained outside the live development budget. The ANAC report offers a narrower comparison: SCML winners used domain-specific heuristics, while an ANL co-winner used reinforcement learning [Aydoğan et al., 2026].

5 Evaluation and Agent Behavior Analysis

All evaluation uses reconstructed per-game *performance rating* P_t. We denote the update rate by V (`ETA` in the code):

$$R_{t+1} = R_t + V(P_t - R_t), \qquad \widehat{P}_t = R_t + \frac{R_{t+1} - R_t}{V}, \qquad V = 0.01. \tag{1}$$

We established V from an uncapped human-facing result: a displayed rating of 1640.8 rose by 13.0 after performance approximately 2940, giving $13.0/(2940 - 1640.8) \simeq 0.0100$. Thus this is an empirical calibration of a linear update model, not a fitted payoff variance. API snapshots supply the pre/post family ratings and game counts; we retain only single-game increments with a unique same-family log match. Rounded or capped UI performance captions are not used as rewards. Calibration

Table 1: Reported within-round comparisons against control lines. Effects are descriptive; bundles and adaptive selection limit causal attribution.

Lever	Family	Effect	Decision
Seat-split acceptance ($\geq$0.30 when they close)	Negotiation	$+50$ to $+230$	adopted
Opponent classification + 4-round anchor	Negotiation	best cells 2443/2536	adopted
Firm proposer line (0.60/0.45, Boulware 0.35)	Bargaining	$+51$ to $+107$	adopted
90/10 last-word commit ($\delta \geq 0.95$)	Bargaining	$+116$ to $+193$	adopted
Disabling the RL menu under overlays	Bargaining	$\sim+100$	tested
Buy only after a seller decline	Persuasion	burns 52%$\rightarrow$38%	adopted
One-strike trust after a burn	Persuasion	$+60$ to $+90$	adopted
Terse categorical messages	Persuasion	$+165/+48$ (two carriers)	adopted
Persuasive text (reveal, deter, evidence, announce)	All	-40 to -180	dropped

precision and timestamp matching are detailed in appendix A.4. The model has an approximately 100-game decay timescale (about 95% adaptation after 300 games), rather than a fixed 100-game window.

Evaluation protocol. Five variants sharing one library ran concurrently, enabling within-round comparisons against a control line. These were live engineering comparisons, not randomized, preregistered trials. Per-game standard deviations were about 450–500: at $N \simeq 160$ per arm the standard error of one mean is 36–40, and of an independent difference about 50–56. Effects near 100 points are therefore suggestive, not automatically significant; we did not correct for multiple comparisons or adaptive selection. The field also drifted: unchanged configurations moved $+30$ to $+130$ between windows. We compared within rounds and sought replication on a second carrier. The terse-offer lever changed from $+118$ to -55 to $+44$ across windows, illustrating this instability.

Competition results and ablations. frodo finished 14th at 2219.8 (top 1: 2594.5), averaged over three families; strongest cells: negotiation $\sim$2530, persuasion $\sim$2280, bargaining $\sim$2140, vs. family means of $\sim$1880–2015 (table 1).

The largest correction was the seat split: 1,172 no-deals in the opponent's-last-word seat averaged 1702 (74% of those games below the field median), while deal outcome bands in that seat averaged 2117–2499. Pooling seats concealed the cost of stubbornness when the opponent controlled the final offer. In the analyzed extraction cells, scores typically plateaued near $R_{\text{opp}} + 800 \approx 2850$, where R_{opp} is opponent rating; this is an empirical pattern, not a universal bound. Accepted extractions of at least 0.85 share often scored only 2200–2300. Opponent-strength weighting means a high payoff against a weak opponent need not earn an exceptional score; rare strong-opponent games could score around 5000. Likewise, $\geq$0.70 shares in *easy* open-horizon cells scored 1775 vs. 2139–2272 for the 0.55–0.70 band: cells where extraction is easy are ones the whole field wins; the percentile rewards beating the field, not the opponent.

5.1 Behavior alignment

Behavior was verified against design on three levels: (i) *legality*: schema validator plus the arena regression of section 2; (ii) *rule firing*: attributed games scanned per outcome band confirm each rule fires in its intended cell (no acceptance below shipped floors); (iii) *live monitoring*: mean performance per family since restart, exposing a bad line within $\sim$1 h.

6 Limitations and reproducibility

These adaptive comparisons cover one changing field; small cells and timestamp matching limit causal inference. Rating inversion assumes $V = 0.01$, calibrated from rounded displays. The retained archive contains 25,257 attributed games versus approximately 27,000 in the historical report, preventing exact replication of all aggregates. Code is available on request.

References

R. Aydoğan, T. Baarslag, T. C. P. Florijn, K. Fujita, C. M. Jonker, and Y. Mohammad. [COMP25] The Automated Negotiating Agents Competition (ANAC) 2025 Challenges and Results. In *Proceedings of the Thirty-Fifth International Joint Conference on Artificial Intelligence, IJCAI-26*, pages 8363–8366. International Joint Conferences on Artificial Intelligence Organization, 2026. doi: 10.24963/ijcai.2026/957. URL https://ijcai.org/proceedings/2026/957. Demo Track.

F. Bianchi, P. J. Chia, M. Yüksekgönül, J. Tagliabue, D. Jurafsky, and J. Zou. How well can LLMs negotiate? NegotiationArena platform and analysis. In *Proceedings of the 41st International Conference on Machine Learning*, volume 235 of *Proceedings of Machine Learning Research*, pages 3935–3951. PMLR, 2024. URL https://proceedings.mlr.press/v235/bianchi24a.html.

P. Faratin, C. Sierra, and N. R. Jennings. Negotiation decision functions for autonomous agents. *Robotics and Autonomous Systems*, 24(3-4):159–182, 1998. doi: 10.1016/S0921-8890(98)00029-3. URL https://doi.org/10.1016/S0921-8890(98)00029-3.

C. Hafer. Communication and standoff. *Games*, 17(1):7, 2026. doi: 10.3390/g17010007. URL https://www.mdpi.com/2073-4336/17/1/7.

A. M. Jakobsen. Coarse Bayesian updating. *The Review of Economic Studies*, 2026. doi: 10.1093/restud/rdag016. URL https://doi.org/10.1093/restud/rdag016. Advance access publication, 18 February 2026.

Y. Pan, S. Zhao, M. Tambe, and Y. Chen. Learning to persuade a biased receiver. *arXiv preprint arXiv:2605.15331*, 2026. doi: 10.48550/arXiv.2605.15331. URL https://arxiv.org/abs/2605.15331.

C. Qian, K. Zhu, J. J. Horton, B. S. Manning, V. Tsai, J. Wexler, and N. Thain. Strategic tradeoffs between humans and AI in multi-agent bargaining. In *Proceedings of the 31st International Conference on Intelligent User Interfaces*, pages 1625–1646. ACM, 2026. doi: 10.1145/3742413.3789078. URL https://doi.org/10.1145/3742413.3789078.

C. Shah, A. Agarwal, K. Garg, and M. Heddaya. LLM Rationalis? measuring bargaining capabilities of AI negotiators. In *First Workshop on Multi-Turn Interactions in Large Language Models at NeurIPS 2025*, 2025. doi: 10.48550/arXiv.2512.13063. URL https://arxiv.org/abs/2512.13063.

E. Shapira, O. Madmon, I. Reinman, S. J. Amouyal, R. Reichart, and M. Tennenholtz. GLEE: A unified framework and benchmark for language-based economic environments. *arXiv preprint arXiv:2410.05254*, 2024. doi: 10.48550/arXiv.2410.05254. URL https://arxiv.org/abs/2410.05254.

J. Shymanski. ChargingBoul: A competitive negotiating agent with novel opponent modeling. *arXiv preprint arXiv:2512.06595*, 2025. doi: 10.48550/arXiv.2512.06595. URL https://arxiv.org/abs/2512.06595.

A Implementation details

This appendix describes the retained implementation, separating its actual rules from motivating analogies. The source files named below are relative to the project root. No language model is called at action time. Parameter values describe the retained code; historical ablations used different override dictionaries.

A.1 Menu learner and terminal reward

Arms and contexts. The menu in `strategy/policies.py` contains four bargaining arms (`base`, `risk`, `hard`, `soft`), five negotiation arms (`base`, `hard`, `soft`, `close`, `greedy1`), and four persuasion arms (`base`, `long`, `thin`, `honest`). Each arm is a dictionary of parameter overrides to a common family policy, rather than a separately trained network. For example, bargaining `soft` sets aspiration 0.55, minimum acceptance 0.22, and Boulware exponent 0.60; `hard` uses 0.72, 0.42, and 0.20.

The exact configuration key is expressed from the agent's own seat. Bargaining includes pot size, own and opponent discounts, horizon and whether it is known, information visibility, and messaging availability. Negotiation includes role, visible valuations, horizon, information, and messaging; persuasion includes role, price, quality prior, high-quality value, horizon, seller information, and message format. Unknown values stay absent. Coarse keys retain role where relevant, horizon class (one round, short ≤ 5, long, or open), and information; bargaining also splits own discount at 0.95. Lookup order is named opponent within exact configuration, exact configuration, coarse context, family, then the shipped `base` policy. Exact/opponent cells require at least 8 total observations; coarse cells require 12. These thresholds are totals over arms, not minima for each arm.

Exact selection rule. Let $D_{c,a}$ be the stored terminal performance scores for arm a in the chosen context c. At every table rebuild, if an arm has no observation, the code chooses uniformly among the least-observed arms. Otherwise *frodo* computes

$$Z_a = X_a + \epsilon_a, \quad X_a \sim \mathrm{Uniform}(D_{c,a}), \quad \epsilon_a \sim \mathcal{N}(0, 80^2), \qquad a^* = \arg\max_a Z_a. \tag{2}$$

The implementation labels this `thompson`, but it samples a smoothed empirical *return*, not a Bayesian posterior of expected reward. Its exploration is heuristic; no posterior-calibration or regret guarantee is asserted. The comparison agent *j-star* uses the same menu and scores with

$$a^* = \arg\max_a \left[\bar{P}_{c,a} + 150 \sqrt{\frac{2\log(\max(N_c, 2))}{n_{c,a}}} \right], \quad n_{c,a} = |D_{c,a}|, \quad N_c = \sum_a n_{c,a}, \tag{3}$$

after the same cold-arm rule. The constants 80 and 150 are in rating points and are the retained deadline-mode settings in `learn.py`. Scores are not clipped or rescaled by pot size. Although a `BINS` constant remains in that file, the sampling function directly draws from score lists; it does not use histogram bins or a Dirichlet model.

Reward, refresh, and overlays. The terminal reward is $r_g = \widehat{P}_g$ from appendix A.4. No intermediate reward, token cost, fairness bonus, or raw-payoff term is added. `learnloop.py` defaults to rebuilding the attribution store and both policy tables every 300 seconds, writing each JSON table atomically. Runtime lookups reload changed tables between actions, so a long game may straddle a rebuild. `GameLog.note_arm` records the latest menu arm for the game; a single terminal reward is assigned to that logged arm, not to every intermediate action. This creates an attribution limitation when menu choices change mid-game.

The exact/opponent pools are built from the same named-game observations. In the retained builder, named observations can consequently enter the family fallback through both pools; these are not independent additional games. The total archive count below counts unique reward rows.

Family-level overrides in `frodo/rl.py` are applied after menu parameters. The enabled opponent-class overlay then retunes relevant parameters before the family policy acts. `menu_off` bypasses menu overrides; in the retained *frodo* snapshot it is enabled for negotiation. The comparison labeled "disabling the RL menu" in the body describes an experiment, not proof that every final family disabled it. Retired state-action learners remain in source for offline experiments and are not the live selection rule described here.

A.2 Opponent classification and behavioral changes

Observable sequence. `strategy/chargingboul.py` extracts opponent bids as utilities to us, rounded to four decimal places. For bargaining, it divides our offered allocation by the current pot. In negotiation, with our private valuation $v > 0$ and opponent price p_i, utility is $(p_i - v)/v$ for us as seller and $(v - p_i)/v$ for us as buyer. These values need not lie in $[0, 1]$. No class is returned until at least three bids are available. The current last-offer handling and the underlying price-history extractor determine which bids enter the sequence; these labels are operational rules rather than estimates of an opponent's latent psychological type.

Statistics and thresholds. For a utility sequence u, split it at $m = \lfloor |u|/2 \rfloor$ into early and late halves. The recursive average-utility index (AUI) increments by one if the late mean exceeds the early mean by 10^{-9}, then recurses on the late half; otherwise it stops. Despite the name, it counts successful mean-improvement comparisons, not the numerical average increment per round. The unique-bid index (UBI) similarly counts recursive increases in the number of distinct bids and is logged for diagnostics. The classification decision itself uses AUI and the early-repeat ratio

$$\rho = 1 - \frac{|\operatorname{unique}(u_1, \ldots, u_m)|}{m}. \tag{4}$$

It returns *hardliner* if AUI ≤ 0, *Boulware* if AUI > 0 and $\rho \geq 0.5$, and *conceder* otherwise. UBI does not determine the class in this short-horizon adaptation. These thresholds differ from the original ChargingBoul procedure [Shymanski, 2025]; they were chosen for GLEE's shorter histories.

Table 2: Parameters set by the class overlay. Empty Boulware bargaining overlay preserves base parameters.

Family	Class	Parameters set
Negotiation	Conceder	`non_closer_hold=True`, share 0.80, acceptance 0.60
Negotiation	Hardliner	`non_closer_hold=True`, share 0.65, acceptance 0.45
Negotiation	Boulware	`non_closer_hold=False`; use the base curve
Bargaining	Conceder	exponent 0.10, aspiration 0.70, minimum acceptance 0.45, snap acceptance 0.60
Bargaining	Hardliner	exponent 0.10, aspiration 0.62, minimum acceptance 0.40, snap acceptance 0.50
Bargaining	Boulware	No parameter changes

Negotiation shares are fractions of the estimated surplus. Holds apply to the non-closer branch; a closer commitment and the initial four-round anchor have precedence in the target-price function. The base surplus floor may further restrict acceptance, so a class's acceptance value is not an unconditional promise to accept. Bargaining uses the time-dependent term $1 - t^{1/\beta}$ (with bounded progress t); $\beta = 0.10$ makes concession late but not identically zero. The last-word commit can take precedence there as well. The classifier runs only if `chargingboul` is enabled, can be disabled with `cb_off`, and does not classify persuasion opponents. The persuasion buyer's separate honest/strategic/spammer model should not be confused with these three bid-pattern classes.

A.3 Text generation and template selection

All outgoing language is selected and filled by Python functions in the family strategy modules. Numerical policy decisions are computed first. Templates receive prices, remaining rounds, known valuations, recommendation decisions, or observed recommendation records. There is no model prompt, decoding temperature, API call, candidate ranking model, or online language optimization.

In negotiation, `_styled_message` first checks `terse_offer`; it can output "Price: 100. Your profit at this price: 20. If we do not agree: 0." when the opponent valuation is known. Otherwise it uses a commitment-style terse sentence. Next come an enabled commitment-revelation template and a non-closer deterrent template. If none applies, `_message` chooses a standard opening, continuation, or last-round sentence. The numerical example illustrates formatting; it is not a measured game transcript. Messages are attached only when the configuration permits them. A stated profit is based on the valuation used by the template, not new external evidence.

In persuasion, the recommendation rule first produces BUY or PASS. If the game requires a binary seller recommendation, the action is the corresponding yes/no decision and no prose is needed. In text mode, the seller renders a recommendation and optionally its observed record. `terse_message` replaces this with "Recommendation: BUY." or "Recommendation: PASS.", followed, when available, by "Record so far: h of n recommended units were high quality." Other flags can add urgency or a first-round statement of the selling scheme. The retained *frodo* persuasion dictionary enables the terse form. Bargaining messages similarly express a computed split and round-dependent rationale; message variants are controlled by parameter flags rather than generated as free text.

The common `trim` function enforces the server's 2,000-character limit (strings longer than that are cut to 1,999 characters). Message experiments compare these fixed styles across live windows. The negative results concern these templates and their interactions with the numerical policy; they do not show that all natural-language persuasion is ineffective.

A.4 Rating calibration, attribution, and archive scope

Establishing V. The retained notes in `ratings.py` record a human-facing before-rating 1640.8, change $+13.0$, and uncapped performance caption approximately 2940. Under the linear update model,

$$\widehat{V} = \frac{13.0}{2940 - 1640.8} = 0.010006\ldots \simeq 0.01. \tag{5}$$

This identifies the effective step size at displayed precision. It is not independent proof of the full server implementation or of a permanently fixed rate. A second recorded result, before-rating 1617.7 and change $+38.6$, is reconstructed as 5477.7 using $V = 0.01$, although its UI caption was approximately 5000. That mismatch motivates using API deltas instead of rounded/capped captions; it is not an independent uncapped calibration point. We preserve the observed evidence rather than asserting more calibration trials than are recorded.

Recovering a game's score. `ratings.py` polls family ratings and counts from `/stats`. For adjacent snapshots in the same agent and family, `reward.py` requires the count to increase by exactly one and computes

$$\widehat{P}_g = R_{\text{before}} + \frac{R_{\text{after}} - R_{\text{before}}}{0.01}. \tag{6}$$

It first matches an unclaimed same-family game with logged completion time in the sample interval; exactly one candidate is required. A second pass widens unmatched intervals by up to 240 seconds on each side to allow delayed completion discovery, again requiring one unclaimed candidate. Matched games are claimed once within that build. Ambiguous or multi-game increments are not assigned a reward. The widened pass is a matching heuristic and can retain timing errors even when a candidate is unique; rejected windows can also introduce selection bias.

Ratings are family-specific, so this procedure does not invert the displayed overall average across three families. Under the assumed update, R_t is an exponential moving average with old weight $(1 - V)^k$ after k games. At $V = 0.01$, $1 - (1 - V)^{100} \simeq 0.634$ and $1 - (1 - V)^{300} \simeq 0.951$. These are decay timescales, not a hard truncation of history. If displayed endpoint ratings were rounded to 0.1, two endpoint errors of at most 0.05 could contribute about 10 points of uncertainty after division by V; using API precision avoids the UI's additional caption cap. A change in the actual V would systematically rescale all reconstructed rewards.

Retained evidence and reproducibility limits. The local `rewards.sqlite` snapshot inspected for this revision contains 25,257 unique attributed games: artemis 5,371; cucu 5,083; frodo 4,946; j-star 4,896; and jshot 4,961. The accepted title and original live-analysis account use the approximate historical total of 27,000. The retained database is smaller and does not establish that exact historical total or reproduce every window-specific aggregate in the submitted report. Reported ranks, effects, and historical cell summaries are retained observations, not newly rerun experiments. Access to the complete historical snapshots and deployment dictionaries would be needed for an exact end-to-end replication.

The main findings should therefore be read as descriptive engineering evidence about seat, reward normalization, and policy simplicity in this field. No universal extraction bound, randomized treatment effect, formal bandit guarantee, or model-wide inability to use messages follows from them.